

FBISE CLASS 9 MATHEMATICS

Chapter 2: Logarithms

Comprehensive Practice Test - FBISE Style

Prepared by Fakhar STEM Sphere

Name:	Roll No.:	Date:
Suggested Time: 2 Hours	Total Marks: 55	Score: /55

General Instructions: Attempt all questions. Show complete working in Section B. Calculators/logarithm tables may be used where permitted by your teacher.

SECTION A - OBJECTIVE

Marks: 15

Q.1 Choose the correct option. Each part carries 1 mark.

- If $\log(x + 3) = \log(15x - 4)$, then x is:
(A) 0.5 (B) 7
(C) 14 (D) 2
- $\log_7(7^{-3}) + \log_2(4^3)$ is:
(A) 3 (B) -3
(C) 0 (D) ± 3
- $\log_b(M + N)$ is:
(A) $\log_b(MN)$ (B) $\log_b M + \log_b N$
(C) Both (A) and (B) (D) None of these
- $\log_b(g^h)$ is:
(A) $g \log_b h$ (B) $\log_b(gh)$
(C) $h \log_b g$ (D) $h \log_g b$
- $\log_b M - \log_b N$ is:
(A) $\frac{\log_b M}{\log_b N}$ (B) $\log_b \left(\frac{M}{N} \right)$
(C) $\log_N M$ (D) $\frac{\log_b N}{\log_b M}$
- $\log_{\sqrt{10}}(100^2)$ is:
(A) 2 (B) 1
(C) 4 (D) 8
- $\log 18$ is:
(A) $3 \log 2 + \log 3$ (B) $\log 2 + 2 \log 3$
(C) $3 \log 3 + 2 \log 2$ (D) $2 \log 3 + 3 \log 2$
- If $\log_{\sqrt{x}} 25 = 4$, then x is:
(A) +5 (B) -5
(C) ± 5 (D) Impossible

SECTION A - OBJECTIVE (Continued)

9. The base of the natural logarithm is:
 (A) Rational number (B) Integer
 (C) Irrational number (D) 10
10. $\log_{\sqrt{b}}(10^4) \div \log_{\sqrt{b}}(10)$ is:
 (A) $\log_{\sqrt{b}}\left(\frac{10^4}{10}\right)$ (B) $\log_{\sqrt{b}}(10^4) - \log_{\sqrt{b}}(10)$
 (C) 4 (D) $\log_{\sqrt{b}}(10^4 - 10)$
11. $5 \log 2 - 2 \log 5$ is:
 (A) $\frac{(\log 2)^5}{(\log 5)^2}$ (B) $\frac{\log 2^5}{\log 5^2}$
 (C) $\log\left(\frac{2^5}{5^2}\right)$ (D) $\frac{5}{2} \log\left(\frac{2}{5}\right)$
12. $\log_y x$ will be equal to:
 (A) $\frac{\log_z x}{\log_y z}$ (B) $\frac{\log_x z}{\log_y z}$
 (C) $\frac{\log_z x}{\log_z y}$ (D) $\frac{\log_z y}{\log_z x}$
13. $\log_b a \times \log_c b$ can be written as:
 (A) $\log_a c$ (B) $\log_c a$
 (C) $\log_a b$ (D) $\log_b c$
14. The relation $y = \log_z x$ implies:
 (A) $x^y = z$ (B) $z^y = x$
 (C) $x^z = y$ (D) $y^z = x$
15. If $a^x = n$, then:
 (A) $a = \log_x n$ (B) $x = \log_n a$
 (C) $x = \log_a n$ (D) $a = \log_n x$

SECTION B - SHORT QUESTIONS

Marks: 40

Q.2 Answer all ten questions. Each question carries 4 marks.

- (i) A radioactive substance decays according to

$$M = 200 \times 10^{-0.05t},$$

where M is the mass in grams and t is the time in years.

- (a) Use logarithms to express t in terms of M .
 (b) Find the time t when the mass M reduces to 100 grams.

- (ii) Bani Gala had a population of 10,000 people in the year 2015. The population has been growing exponentially at a rate of 2.5% per year. Using the exponential growth formula

$$P(t) = P_0 e^{rt},$$

apply laws of logarithms to determine the year when the population reaches 25,000.

SECTION B - SHORT QUESTIONS (Continued)

- (iii) The population of a colony is growing according to

$$P = 2500e^{0.09t},$$

where P is the population and t is the time in years.

- (a) Use logarithms to express t in terms of P .
(b) Calculate the time t when the population increases to 10,000.
- (iv) An earthquake that occurred in Pakistan in 2005 measured 7.6 on the Richter scale, while an earthquake in China measured 8.2. How many times stronger was the earthquake in China than the earthquake in Pakistan? Use

$$M = \log \left(\frac{I}{I_0} \right).$$

- (v) Starting with 500 users, a startup experiences continuous growth at a rate of 8% per year. Determine the number of years it takes the company to reach 3,000 users using

$$N(t) = N_0e^{rt},$$

where $N(t)$ is the number of users after time t , N_0 is the initial number of users, and r is the growth rate.

- (vi) An earthquake in 1920 measured about 8.5 on the Richter scale and another earthquake in 1923 measured 7.8. How many times stronger was the 1920 earthquake than the 1923 earthquake? Use

$$M = \log \left(\frac{I}{I_0} \right).$$

SECTION B - SHORT QUESTIONS (Continued)

(vii) Find the number of digits in 5^{50} .

(viii) Use laws of logarithms to evaluate:

$$\frac{\sqrt[3]{8.59} \times (55.6)^2}{2.51 \times \sqrt{2.12}}.$$

(ix) Given

$$A = A_0 e^{-kd}.$$

If $k = 2$, what should be the value of d to make $A = \frac{A_0}{2}$?

(x) Evaluate:

$$\sqrt[3]{\frac{0.07921 \times (18.99)^2}{(5.79)^4 \times 0.9474}}.$$

End of Test - Best of Luck!